

Exam-wise Notes — DSSSB TGT Computer Science (DOE)

Languages (Java/.NET/Python) • Web & Front-end • Graphics • MIS/E-Commerce/Mobile • Linux • IT Fundamentals
• Pandas

How to read this exam-wise PDF

Covers the CS/IT topics **unique to DSSSB TGT CS (DOE syllabus, the governing body)** that the shared umbrella did not teach — the DOE heads beyond the shared core.

Depth is **graduation/teaching level** (what DSSSB TGT tests): concept clarity + terminology, not GATE-depth derivations.

■ **Green box** = how DSSSB / govt CS papers test the topic. ■ **Purple box** = scope note.

Shared subjects (OS, DBMS, CN, DS, programming basics, digital logic) stay in umbrella Modules 1-6. This PDF adds only DSSSB-specific heads. AI/ML/Emerging-tech is in the separate shared **Emerging Tech & Security** PDF.

0 What this PDF covers

DSSSB DOE's syllabus is the broadest of your three — 32 heads. Its shared-core heads (C/DS, DBMS, OS, CN, DAA, digital electronics, computer architecture, software engineering) are already in your umbrella. This PDF supplies the remaining DOE-specific CS heads:

- **Part A — IT Fundamentals & PC Software** (also DSSSB NDMC).
- **Part B — Programming languages:** Python (+ Pandas), Java, .NET / C#.
- **Part C — Web & Internet technologies:** HTML/CSS/XML, front-end tools, web servers, client-server.
- **Part D — Computer Graphics & Multimedia.**
- **Part E — Applied IT:** MIS, E-Commerce, Mobile Computing, Linux Environment.

■ Tested pattern & focus (PSU / Govt CS-IT-Systems)

DSSSB TGT asks **direct concept & terminology MCQs** plus some **code/output** reading (mostly Python). No heavy derivations.

Highest-yield: **Python basics & data types**, **OOP-in-Java** terms, **HTML tags & CSS**, **MIS/e-commerce definitions**, **Linux commands**, and **graphics primitives**.

PART A IT Fundamentals & PC Software

A.1 Computer basics

A **computer** is an electronic device that takes **input**, **processes** it per stored instructions, produces **output**, and can **store** data. Generations: vacuum tubes → transistors → ICs → microprocessors → AI/ULSI. (Hardware/CPU/memory internals are in umbrella Module 5.)

- **Software types:** **system software** (OS, utilities, device drivers, language translators) vs **application software** (word processors, browsers) vs **utility software**. **Generic** (off-the-shelf) vs **specific/custom** (bespoke).
- **PC software / office tools:** word processing, spreadsheets (cells, formulas, functions), presentations, DBMS front-ends.
- **File management basics:** files, folders, paths, extensions.

A.2 Number representation recap

Data is stored in **bits/bytes** (1 byte = 8 bits). Units: KB, MB, GB, TB, PB. Character encoding: **ASCII** (7/8-bit), **Unicode** (multilingual). Detailed number-system conversions are in umbrella Module 5.

■ Tested pattern & focus (PSU / Govt CS-IT-Systems)

MCQs: **system vs application software** classification, **generations** of computers, **memory units** order (KB < MB < GB < TB), and **ASCII vs Unicode**.

PART B Programming Languages

B.1 Python (DSSSB's primary language)

Python is a **high-level**, **interpreted**, **dynamically-typed** language. DSSSB (both DOE & NDMC) tests its basics heavily.

- **Tokens:** keywords, identifiers, literals, operators, punctuators. **Indentation** defines blocks (no braces).
- **Data types:** **numbers** (int, float, complex), **bool**, **sequences** (str, list, tuple), **set**, **mapping** (dict), None.
- **Mutable** (list, dict, set) vs **immutable** (int, float, str, tuple) — a favourite MCQ.
- **Control flow:** if/elif/else, for, while. **Functions:** def, arguments (positional/default/keyword), return, scope (local/global), lambda.
- **Files:** open modes (r/w/a/r+), read()/readline()/readlines(), write()/writelines(), with-clause; text vs binary vs CSV.
- **Built-ins commonly tested:** len(), type(), range(), and dict methods keys()/values()/items()/get()/update().

■ Python facts exams love

Mutable vs immutable (list mutable, tuple immutable); **list vs tuple vs set vs dict** differences; **indentation** = block structure; output of a small loop/slice. Slicing: **s[start:stop:step]**, stop is exclusive.

B.2 Pandas (data handling — DSSSB NDMC)

- **Series** — a 1-D labelled array. **DataFrame** — a 2-D labelled table (rows + columns).
- Create from dict/list/CSV; access via labels/index; **head()/tail()**, boolean indexing, add/select/delete/rename columns.
- **Matplotlib** — basic plotting library paired with Pandas.

B.3 Java

- **Platform-independent** via the **JVM**: source (.java) → **bytecode** (.class) by the compiler → run by the JVM ('write once, run anywhere').
- Purely **object-oriented** (everything in classes). Supports the OOP pillars (see umbrella Module 4): encapsulation, inheritance, polymorphism, abstraction.

- **No multiple inheritance of classes** — uses **interfaces** instead (avoids the diamond problem). Automatic **garbage collection**.
- Key terms: JDK (dev kit), JRE (runtime), JVM (virtual machine); class, object, method, constructor, package.

B.4 .NET / C#

- **.NET** is Microsoft's framework; **C#** is its main language. Code compiles to **CIL/MSIL**, run by the **CLR** (Common Language Runtime) — conceptually like the JVM.
- **ASP.NET** — for building web applications (DSSSB NDMC mentions ASP). Supports OOP, is language-interoperable (VB.NET, C#, F#).

■ Tested pattern & focus (PSU / Govt CS-IT-Systems)

Language MCQs: **compiled vs interpreted** (Java bytecode+JVM, Python interpreted, C compiled); **Python mutable/immutable** & output; **Java = JVM/bytecode/platform-independent, interfaces not multiple inheritance**; **.NET = CLR/C#**.

PART C Web & Internet Technologies

C.1 How the web works (client-server)

A **web browser** (client) requests pages from a **web server** over **HTTP/HTTPS**; the server returns HTML which the browser **renders**. A **URL** = protocol + domain + path; **DNS** resolves the domain to an IP (see umbrella Module 1). **Static** pages are fixed files; **dynamic** pages are generated per request (server-side).

C.2 HTML

- **HTML** (HyperText Markup Language) structures a page with **tags/elements**. Basic skeleton:
`<html><head><body>`.
- Common tags: headings `<h1>-<h6>`, paragraph `<p>`, links `<a href>`, images `<img>`, lists `<ul>/<ol>/<li>`, tables `<table><tr><td>`, forms `<form><input>`.
- **DHTML** = dynamic HTML (HTML + CSS + JavaScript + DOM). **DOM** = Document Object Model, the tree representation of a page that scripts manipulate.

C.3 CSS & XML

- **CSS** (Cascading Style Sheets) — controls presentation (colour, layout, fonts). Applied **inline**, **internal**, or **external**; the **box model** = content + padding + border + margin.
- **XML** (eXtensible Markup Language) — stores/transport data with custom tags; **self-describing**, needs well-formedness; structure defined by a **DTD** or schema. HTML describes *presentation*, XML describes *data*.
- **Front-end tools / scripting**: JavaScript (client-side interactivity); DSSSB NDMC also lists ASP (server-side).

■ Tested pattern & focus (PSU / Govt CS-IT-Systems)

Web MCQs: **HTML tag purposes**, **HTML vs XML** (presentation vs data), **CSS ways to apply** & the box model, **DOM** definition, and **client-side vs server-side**.

PART D Computer Graphics & Multimedia

- **Pixel** = smallest picture element; **resolution** = pixels (width×height). **Raster/bitmap** graphics (pixel grid: JPEG, PNG) vs **vector** graphics (math-defined shapes: SVG — scalable without quality loss).
- **Scan conversion** — drawing primitives on a raster: line algorithms (**DDA**, **Bresenham**), circle algorithms.
- **2D transformations** — translation, rotation, scaling, reflection, shearing (via matrices).
- **Clipping** — keeping only the part of an object inside a window (e.g. Cohen-Sutherland line clipping).
- **Colour models**: **RGB** (additive, screens) vs **CMYK** (subtractive, printing).
- **Multimedia** = text + audio + image + video + animation; common formats (JPEG/PNG/GIF, MP3, MP4).

■ Tested pattern & focus (PSU / Govt CS-IT-Systems)

Graphics MCQs: **raster vs vector**, **pixel/resolution**, **line-drawing algorithms** (DDA/Bresenham) names, **RGB vs CMYK**, and **types of transformations**.

E.1 Management Information System (MIS)

- **MIS** — a system that collects, processes, stores and delivers information to support **managerial decision-making**.
- Related systems: **TPS** (transaction processing), **DSS** (decision support), **ESS** (executive support), **ERP** (enterprise resource planning).
- Data → information → knowledge hierarchy; qualities of good information (accurate, timely, relevant, complete).

E.2 E-Commerce

- **E-commerce** = buying/selling goods & services online. Models: **B2B** (business-business), **B2C** (business-consumer), **C2C** (consumer-consumer), **C2B**.
- **Payment**: gateways, wallets, UPI, cards; secured by **SSL/TLS** and **digital signatures** (see IBPS security PDF).
- **EDI** (Electronic Data Interchange) — structured business-document exchange.

E.3 Mobile Computing

- Computing on the move via wireless networks. Generations: **1G**→**2G**→**3G**→**4G**→**5G** (analog voice → digital → data → broadband → ultra-low-latency).
- Concepts: **cellular networks** (cells, base stations, handoff), GSM/CDMA, mobile OS (Android/iOS), Wi-Fi/Bluetooth.

E.4 Linux Environment

- **Linux** — open-source, multi-user, multitasking OS; the **kernel** + **shell** (command interpreter) + file system.
- **File system** is a single tree from root **/**; everything is a file. Case-sensitive.
- **Common commands**: ls (list), cd (change dir), pwd (path), mkdir, rm, cp, mv, cat, grep (search), chmod (permissions), man (help), ps/kill (processes).
- **Permissions**: read/write/execute for owner/group/others (e.g. chmod 755).

■ Tested pattern & focus (PSU / Govt CS-IT-Systems)

Applied-IT MCQs: **MIS vs DSS vs TPS**, **e-commerce models** (B2B/B2C/C2C), **mobile generations** (what 4G/5G add), and **Linux commands + chmod permissions**. These are pure recall — easy marks.

■ Coverage note / warning

Everything here is **teaching/graduation level** — DSSSB rewards breadth of clear definitions, not depth. Don't over-invest; the marks are in terminology and classification, not derivations.

Rapid recap — DSSSB exam-specific

Area	Nail these
IT fundamentals	System vs application software; computer generations; memory units KB<MB<GB<TB; ASCII vs Unicode
Python	Interpreted, indentation=blocks; mutable(list/dict/set) vs immutable(int/str/tuple); slicing s[start:stop:step]; dict methods; file modes
Pandas	Series(1-D) vs DataFrame(2-D); create from dict/CSV; head/tail, boolean indexing; Matplotlib
Java	.java→bytecode→JVM (platform-independent); pure OOP; interfaces not multiple inheritance; JDK/JRE/JVM; garbage collection
.NET	C# on CLR; CIL/MSIL bytecode; ASP.NET for web; language-interoperable
Web	HTML(structure)/CSS(style, box model)/XML(data); DOM=page tree; DHTML; client vs server side; HTML vs XML
Graphics	Raster/bitmap vs vector; pixel/resolution; DDA/Bresenham lines; 2D transforms; RGB(screen) vs CMYK(print)
Applied IT	MIS/DSS/TPS/ERP; e-commerce B2B/B2C/C2C; mobile 1G-5G; Linux commands (ls/cd/grep/chmod), permissions rwx

✓ Where this sits

This completes CS/IT coverage for **DSSSB TGT CS (DOE)**: umbrella Modules 1-6 (shared core) + this PDF's DSSSB-specific heads. (AI/ML & emerging tech are in the shared Emerging Tech & Security PDF, next.)

All three target exams are now fully covered on CS/IT. Remaining plan items: the shared **Emerging Tech & Security** PDF, future **library-filler** modules (Distributed Systems, deeper Web, standalone Cryptography, AI/ML), and the **Master Topic Index**. Q&A sets follow your walkthroughs.